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Study on the effect of carbamates on *Atherigona soccata* in maize fields of Madhepura, Bihar

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Abstract- The sorghum shoot fly (Atherigona soccata) is a significant pest affecting maize crops in Bihar, causing considerable yield losses. Carbamate insecticides are commonly used to control pest populations, but their impact on A. soccata in maize fields remains underexplored. This study investigates the mortality and morphological deformities induced by different carbamate insecticides on A. soccata populations in Bihar maize fields. Field and laboratory experiments were conducted to analyze mortality rates, developmental abnormalities, and behavioral changes in response to carbamate exposure. The results indicate that carbamate treatments effectively increased mortality rates while also causing significant deformities, such as incomplete molting, wing deformities, and impaired locomotion in surviving individuals. The findings contribute to improved pest management strategies while considering potential environmental and ecological risks.

Key words: ***Atherigona soccata*****, Carbamate Insecticides, Mortality, Deformity, Maize fields, Pest Management**

INTRODUCTION

Bihar is one of the leading maize-producing states in India, contributing significantly to the country's overall maize production. Maize, also known as corn, is grown extensively in Bihar due to its favorable agro-climatic conditions, fertile soil, and availability of water resources. The key districts involved in maize production in Bihar includes Kishanganj, Purnia, Katihar, Araria, Begusarai, Saharsa, Madhepura, Bhagalpur. The Seemanchal region (comprising Purnia, Katihar, Kishanganj, and Araria) is particularly known as the maize hub of Bihar. Bihar grows maize in all three agricultural seasons such as Rabi (Winter) Maize - Grown from October-November to March-April, this is the most productive season for maize in Bihar due to favorable weather conditions. Kharif (Monsoon) Maize - Sown in June-July and harvested by September-October. However, this season faces challenges like floods and pest attacks. Zaid (Summer) Maize - Grown from February-March to June-July, it is mostly dependent on irrigation.

Bihar has one of the highest maize yields in India, with an average productivity of 4-5 tonnes per hectare, which is much higher than the national average. In some regions, hybrid varieties yield up to 8-10 tonnes per hectare.[1,3] Maize cultivation in Bihar has witnessed remarkable growth due to the adoption of modern farming techniques, government support, and favorable agro-climatic conditions. With proper infrastructure, storage facilities, and stable market prices, Bihar has the potential to become a maize export hub in India.

However, pest infestations, particularly by *Atherigona soccata* (sorghum shoot fly), have been a major challenge in maize production. Traditional pest management relies

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heavily on chemical insecticides, among which carbamates are widely used due to their effectiveness in inhibiting acetylcholinesterase (AChE) in insect nervous systems. This study aims to evaluate the impact of carbamate insecticides on the mortality and deformities of A. soccata under field and laboratory conditions.

MATERIALS & METHODS

Study Area

Maize was gathered from Madhepura town's neighboring villages. Farmers from many villages, including Sahugarh, Gamharia, and Sabaila, were approached. To evaluate the mycoflora, seeds that were typically kept in earthen pots, "Kothi" earthen storage boxes, and containers. Additionally, the seeds' storage times were recorded. In sterile, airtight plastic bags, each type of seed was gathered separately. Every sample that was collected and contained plastic bags was properly labelled. The study is carried out in the Madhepura District, which is situated between latitudes 25°31' and 26°20' and longitudes 86°36' to 87°07'. The district has temperatures ranging from 35 to 45 degrees Celsius at its highest point and 7-9 degrees Celsius at its lowest point. On average, the district receives 1,300 mm^3 of rainfall. The experiment followed a randomized block design with control and treatment plots.



A



B



B

Fig. 1 - Map showing different study area
A. Sahugarh, B. Gamharia, C. Sabaila
Source Google map

Statistical Analysis

The results were statistically analyzed using ANOVA to determine significant differences among treatments. Regression analysis was used to establish dose-response relationships. ANOVA will help determine whether there are statistically significant differences in the means of multiple treatment groups.

In the kharif season of 2023, a field experiment was set up to investigate the effectiveness of treating seeds with carbamates to control sorghum shoot flies. This was done in contrast to the suggested practice, which is to spray carbamates at a rate of 650 ml/ha (T2), while the untreated plot was designated as T1. The la provided treated seed samples that had been treated with the test pesticide. Control plots were used to compare the therapy. All of the suggested measures, including fertiliser application and irrigation, were used to raise the crop. The percentage of dead hearts caused by sorghum shoot flies was noted at 14, 21, and 28 days following crop germination. For each therapy, data on dead heart development was collected from ten distinct locations. Deformities in surviving larvae and pupae were also observed and documented.

RESULTS & DISCUSSION

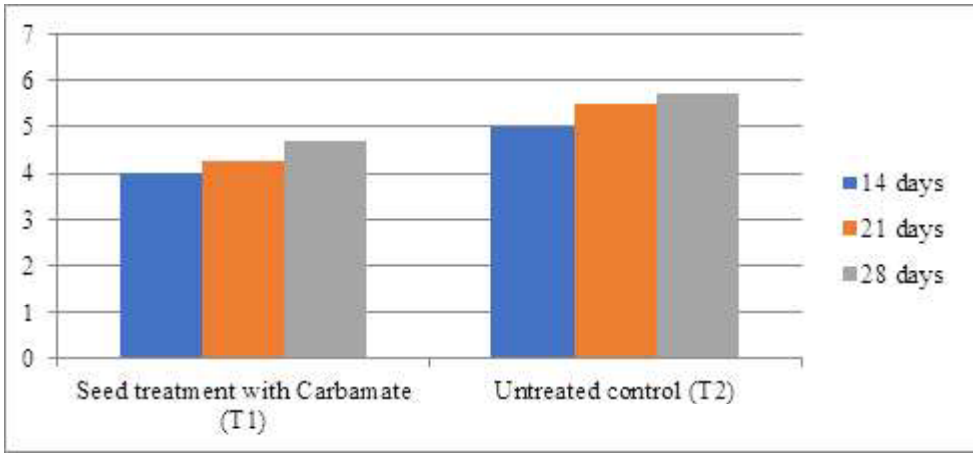
Carbamate insecticides significantly increased mortality rates in A. soccata populations. Sublethal exposure to carbamates induced severe developmental abnormalities,

including, Incomplete molting leading to larval mortality; Wing deformities, impairing flight capability; Malformations in antennae and legs, affecting sensory perception and locomotion; reduced feeding behavior, leading to weakened individuals unable to reproduce effectively. Compared to biological control methods such as parasitoids and neem-based bio-pesticides, carbamates provided rapid pest suppression but posed risks of resistance development and non-target effects. The study suggests integrating chemical and biological control methods for sustainable pest management. Carbamate residues in wheat fields may lead to soil contamination and potential toxicity to non-target organisms, including pollinators and natural enemies of pests. Future studies should focus on assessing residue degradation rates and safer application strategies.

According to the experimental data, there were notable variations in the number of dead hearts produced by shoot flies in sorghum across all treatments at 14, 21, and 28 days following crop germination as compared to the control. Fourteen days after crop germination, 4.01 percent dead heart development was noted in treatment T1 (Table 1). In the treatment T1, dead heart formation was 4.25% and 4.68% in 21 and 28 days after crop germination respectively. At 14, 21, and 28 days following crop germination, the experimental results showed notable variations in the dead heart production by shoot flies in sorghum across all treatments. Fourteen days after crop germination, 3.28 percent dead heart development was seen in treatment T1. Twenty-one days after crop germination, the dead heart formation was 3.93 percent in treatment T1, whereas it was 3.59 percent in treatment T1.

Table 1 - Per cent dead hearts formed due to sorghum shoot fly

Treatment	Per cent dead hearts formed due to sorghum shoot fly after		
	14 days	21 days	28 days
Seed treatment with Carbamate (T_1)	4.01	4.25	4.68
Untreated control (T_2)	5.02	5.50	5.74



Graph 1 - Showing Per cent dead hearts formed due to sorghum shoot fly

In an study It was obtained by Balikai (2011)[4] that, the seed treatment with thiamethoxam 70 WS @ 3 g/kg seeds proved highly effective against shoot fly and significantly superior over rest of the treatments by recording 5.2 per cent dead hearts. Ten millilitres of Slayer 30 FS (thiamethoxam) per kilogramme of seed was applied to the seed. For the control of shoot flies in sorghum, thiamethoxam 30 FS seed treatment at 10 ml/kg seed was shown to be more successful than spraying malathion at 625 ml/ha and control; however, thiamethoxam seed treatment was ineffective for controlling maize borer in sorghum.[5]

CONCLUSION

The study highlights the effectiveness of carbamate insecticides in controlling A. soccata in Bihar wheat fields. However, the observed deformities raise concerns about sublethal exposure and long-term ecological impacts. Using recommended doses to minimize resistance development and non-target effects. Combining chemical control with

biological alternatives to ensure sustainable wheat production. Conducting further research on carbamate residue persistence in soil and water systems.

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